

Rethinking Heuristics, Are We Heading the Right Way? A Case Study on Heuristic Evaluation in Action

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Abstract

Three decades into the evolution of Heuristic Evaluation (HE), we must ask: are we still on the right path? This study critiques the current trajectory of HE research, arguing it has drifted from its original framework and lacks a clear understanding of real-world applications. Through a case study involving in-depth interviews and actual observations of UX practitioners in a large-scale organization, we uncover the raw, unvarnished practice of HE. Our findings reveal that practitioners often rely on unstructured, experience-based, single-evaluator approaches, diverging from the structured, multi-evaluator methods advocated in recent literature. Additionally, we highlight the overlooked influence of individual, task-related, and organizational factors on HE practice. HE, despite remaining to be a useful tool for UX practitioners, calls for a change in perspective. We challenge the HCI community to rethink HE - to acknowledge that maybe, we need to pivot our approach.

CCS Concepts

• Human-computer interaction (HCI); • HCI design and evaluation methods; • Heuristic evaluation;

Keywords

heuristic evaluation, case study, practical application

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1 Introduction

Heuristic Evaluation (HE) has long been a cornerstone in the field of human-computer interaction (HCI) for its pragmatic approach to usability assessment. It is one of the most utilized and probably the most well-known method for usability evaluation both in research and in practice [2, 8]. In the past few decades, literature has been flooded by studies attempting to redefine and restructure

HE, making it more suitable for a wide range of practical applications. However, as this area of research continues to race forward, it seems that it is heading towards a direction different from where it was once grounded.

HE, formally introduced by Nielsen and Molich in 1990 [11], is a technique originally used to assess the usability of user interfaces based on a set of heuristics (i.e., good practices), which Nielsen refers to as 'broad rules of thumb' that describes what a good interface should be [14]. This is considered as a 'discounted' usability inspection method as it requires less formal procedures, making it more time-efficient and cost effective [13]. Due to its simplicity and perceived efficiency, HE is widely adopted in the industry. Earlier research has shown that 75% of the total usability problems can be identified through HE, involving only three to five usability experts, which is more efficient than formal user testing [12]. However, the credibility and effectiveness of HE in real-world settings have been subjects of debate due to the inherent subjectivity and lack of rigor [1, 6]. As a result, ongoing efforts to examine and continuously refine HE, usually through developing more structured and reliable procedures, are extensively documented in literature [1, 6, 15, 16, 18, 19]. This includes developing more accurate methods of reporting usability issues, creating specific guidelines and checklists, and incorporating quantitative measures into heuristic assessments [4, 15]. This evolution in HE practices represents a shift toward a more robust and comprehensive approach to usability evaluation, with the goal of bridging the gap between theory and practice.

For us, this shift of focus prompts the question, "Aren't we veering off course?" We find it amusing that people are trying to make HE more systematic and objective when the inherent meaning of the term 'heuristic' refers to the exact opposite. Heuristics, in its simplest definition, are mental shortcuts that we use to speed up the process of decision making [7]. In reference with this definition, HE was first introduced as a tool leveraging heuristics, or mental shortcuts, to evaluate system usability in a more efficient and intuitive manner. However, looking at some of the recently published research in HE, most of them seem to suggest ways to follow a structured set of 'heuristics' and ways to quantify it, increasing its complexity and rigor. It then results to some methods that can hardly be called 'discounted' or 'simple' anymore. While the interest in HE is growing, the word 'heuristic' in HE seemed to have lost its meaning.

Another thing that we have noticed is that research on HE seems to overlook the importance of the task environment when investigating HE [15]. It is a misconception to view the evaluator and

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the environment as separate entities. The effectiveness of HE is intrinsically tied to the environment where it is applied – they are interdependent, not isolated. The context to where the evaluation occurs should be investigated as much as the evaluation procedure itself. Since organizations are embedded in dynamic context where evaluators' judgment and decisions are influenced by various environmental factors, expanding our view of HE is necessary.

Despite limited research on the role of the task environment, there seems to be a prevailing presumption that a systematic approach to HE is most effective in practice. We contend that this belief overlooks a fundamental aspect: the reality of HE's application in the practical setting. Much of the past research has been anchored in theory, conducted in controlled, hypothetical environments that do not fully capture the complexities faced in actual practice. This discrepancy between idealized methods and actual execution invites us to rethink our assumptions about what truly makes HE effective. Year after year, we see new guidelines and best practices proposed, offering 'ideal' implementations of HE [15]. But the question remains, are they applicable to real-world situations?

Our research delves into the real-world practices of HE through a case study at a large-scale mobile technology company. Drawing on the insights of seasoned UX researchers, this study provides a rare and comprehensive perspective, given the scarcity of case studies within major corporations. Here, we aim to be a catalyst for a profound shift in understanding HE in real-world contexts, challenging and enriching existing paradigms. Ultimately, our goal is to do more than just inform the HCI community; we aim to stir the pot, challenging long-held notions, and urge a rethinking of HE that truly resonates with reality.

2 Case Study Description

This case study explores the application of HE in an industrial context, identifying factors influencing its practice at both micro and macro levels. To achieve this, we conducted in-depth, one-on-one interviews with experienced UX professionals who regularly perform HE in their work environments. These interviews provided insights into their processes, tactics, challenges, and perspectives. Complementing this, we observed HE practices in action, focusing on workplace behaviors and group dynamics. Below, we detail the study context, participants, data collection, and analysis.

2.1 Study Context

The study was conducted at a large international company producing consumer devices globally. The organization follows a hierarchical structure, with employees and executives further stratified by career level. This research focused on the Customer Experience (CX) research department, a centralized team tasked with generating insights to improve products and inform decisions. At the time, the department comprised over 100 UX professionals with diverse expertise and experience, organized into specialized teams led by team leaders. Most researchers worked on specific products, engaging in evaluative activities throughout the development cycle.

2.2 Participants

Ten UX professionals, including five team leaders, were selected through purposive sampling to participate in the interviews. The

participants had an average of 9.7 years of UX research experience ($SD = 3.28$, minimum = 5 years), with ages ranging from their 30s to 50s. All were seasoned professionals who had conducted extensive quantitative and qualitative research, including HE and user testing, in diverse contexts.

2.3 Data Collection and Analysis

We employed semi-structured qualitative interviews [5], enabling a guided yet flexible conversation to capture participants' viewpoints comprehensively. The interviews covered topics such as the UX research process, how and when HEs are conducted, and their perceived advantages, challenges, and limitations. While a predefined guide was used, the interview format allowed for rephrasing, omitting, or adding questions based on participants' responses. Each in-person interview lasted 30–50 minutes, was recorded, and transcribed for analysis.

Data collection and thematic analysis [10] were performed iteratively until theoretical saturation was achieved. The interview transcriptions were segmented into meaningful units, each representing a single concept, and then coded [3]. Microsoft Excel was used to document this coded data, from which relevant themes were subsequently identified.

As part of the data collection, direct observations of UX practitioners' behaviors and performance within the study context were made. Rather than using specific metrics or criteria, we focused on gathering contextual information and observing group behaviors to gain a comprehensive overview of the case. Additionally, in-house policies, previous evaluation reports, and organizational documents were reviewed. Observations were documented using either a memo app or traditional pen-and-paper methods. This triangulation of methods provided a comprehensive view of HE practices within the organization and supplemented the interpretation of the interview findings.

3 Findings

We present our main findings which are categorized into three subsections: general HE practices, factors influencing HE practice, and UX practitioners' perspective on HE. The emergent themes are elucidated through quotations from the interviewees, each accompanied by a unique identifier for the respondent and is labeled either 'TL' or 'TM' to distinguish between the insights of team leaders and team members, respectively. Note that all the interviewees' statements quoted in this case study are translated from Korean to English.

3.1 General HE Practices

Findings reveal that most UX practitioners adhere to three key HE practices: (1) unstructured approaches, (2) a posteriori evaluations, and (3) single-expert evaluations. Moving away from strict guidelines, practitioners increasingly rely on intuition and personal experience, allowing for greater flexibility in addressing unique design challenges. HE is often conducted by a single expert, leading to quicker and more focused decision-making. These adaptive, experience-driven practices signify a shift from traditional methods, reshaping the standards of HE in dynamic, real-world settings.

3.1.1 Less structured approach to HE. All the participants reported that most, if not all, of the UX practitioners in the team typically adopt a less structured approach to HE. While documented guidelines and heuristics for evaluations were once in place, they are rarely, if ever, used in practice. This shift away from structured checklists and templates, as several participants noted, is largely due to practitioners growing too familiar with the process. As one team leader put it, *“We made a checklist of the things that we need to assess, but as you know these checklists can be so long. So, we only use these checklists in the beginning but when we eventually got used to it, we just do it on our own, usually based on experience.”* (TL-4). It is crucial to clarify that, in this statement and the subsequent ones, the term “checklists” pertains to lists of heuristics or requirements employed for evaluation, such as the well-known 10 usability heuristics of Jakob Nielsen [12].

This shift is largely driven by the rapid pace of product updates and concept changes, which render static checklists impractical: *“Creating checklists is a never-ending process. Products change so much that we stop using them.”* (TM-1). Practitioners also find structured methods cumbersome and overly generic: *“Using checklists is time-consuming and unnatural. Usability issues will always emerge that aren’t covered by these lists.”* (TM-5). This goes to show that in dynamic sectors like tech, such tools fall short, leading practitioners to rely more on their intuition and experience.

3.1.2 A posteriori evaluation: expert knowledge and experience goal. The original HE concept involves a top-down approach, using a set of heuristics based on a priori knowledge of what constitutes a good interface as the criteria for evaluation [14]. However, our observations reveal a contrasting approach in actual practice. UX practitioners increasingly favor a bottom-up method, grounded by their posterior knowledge and the product’s experience goal rather than fixed heuristics. Posterior knowledge involves the evaluators’ expert knowledge and prior experience with the product. This tendency to rely on experiential knowledge was reflected in TM-5’s statement, *“[...] as time goes by, history piles up, and because of that, instead of going one-by-one at each test case, I already got a lot of information about that product, so I tend to focus on what I know is critical.”* In addition, one participant pointed out that *“we also refer to findings from big data analysis, voice of customer (VOC) feedback, findings from previous studies, and etc.”* (TL-5). This suggests that evaluators collect information from various sources directly and indirectly, and use it as a basis for their evaluation.

Many UX practitioners also take into account the experience goal of the product during HE. The experience goal refers to the kind of experience or emotion the users are supposed to have or feel when using the product [9]. In essence, it reflects the intention of the experience designers. According to the participants, identifying the goal of the product or feature is the first step to conducting HE. Instead of asking questions like: Is this less error prone? Is it more visible? Evaluators tend to think more about its conceptual properties and ask questions such as: Why is this feature made? When would the users use this? Why would the users use this? This would give the evaluator a clear picture of the product purpose. Then, the evaluators would identify the main tasks and relevant use case scenarios and use these scenarios to carry out HE. The objective of HE here is to assess whether the goal is being met

and to confirm that there are no usability issues that interfere with its achievement. While this approach can be seen as subjective, participants argue that HE is about prioritizing usability issues in line with the product’s goals rather than addressing every potential problem. This trade-off reflects the constraints of real-world UX practices, where time and resources are limited, and evaluations must align with broader objectives.

3.1.3 Single expert evaluation. HE is traditionally performed by multiple evaluators to reduce bias and errors [6]. However, in practice, single-expert evaluations are the norm, mostly due to efficiency and resource constraints. As one team leader noted, *“A single person evaluates most features, which can result in missed issues, but involving multiple evaluators isn’t always feasible.”* (TL-2). While cross-evaluations are ideal, they are typically reserved for critical projects or issues flagged as high-priority. In such cases, the depth and rigor of multiple evaluations are deemed essential. This oscillation between single and multiple evaluations is not arbitrary, but rather it is a calculated response to the demands of the context in which HE is conducted. This duality speaks volumes about the nuanced role of group dynamics in HE, illustrating that the best approach is often contingent on the specific challenges and stakes of each situation.

Interestingly, social dynamics also seems to play a role in the effectiveness of cross-evaluations. Subtle yet impactful factors, such as not wanting to overshadow the work of the main evaluator, prioritizing one’s own workload, or avoiding conflicts due to differing opinions, further complicate the process. These unspoken social rules seems to significantly influence the outcome and the process of conducting of the HE.

3.2 Factors Influencing HE practice

In the previous section, we enumerated and explained how HE is being implemented in the practical setting. Now, we turn our focus to the multitude of factors that shape these practices. Based on our analysis, there are various factors that influence the evaluators’ judgments and ability to perform HE. These factors range from individual-related factors to larger scale environmental factors [Figure 1]. The results of the analysis have shown that even organizational culture, social dynamics and performance appraisals have a strong impact on the evaluator’s judgment and behavior.

3.2.1 Individual-related Factors. In the intricate process of HE, the unique traits of each evaluator - their knowledge, expertise, and personality - are not just influential but can be pivotal: *“individual characteristics such as attitude and aptitude affect the effectiveness of HEs.”* (TL-1). It was found that experienced evaluators tend to identify specific issues, while less experienced ones focus on general aspects of usability. This findings suggest that an evaluator’s depth of experience and breadth of knowledge could highly influence the quality of HE, either enhancing or undermining its outcomes.

The reliance on posterior knowledge of most experts, stemming from prior experience and other sources such as big data or customer feedback, further highlights the importance of a well-rounded background in evaluators. TL-1’s remark, *“the evaluator was not able to capture a critical usability issue, which was later identified via customer feedback, simply because the evaluator never had a similar*

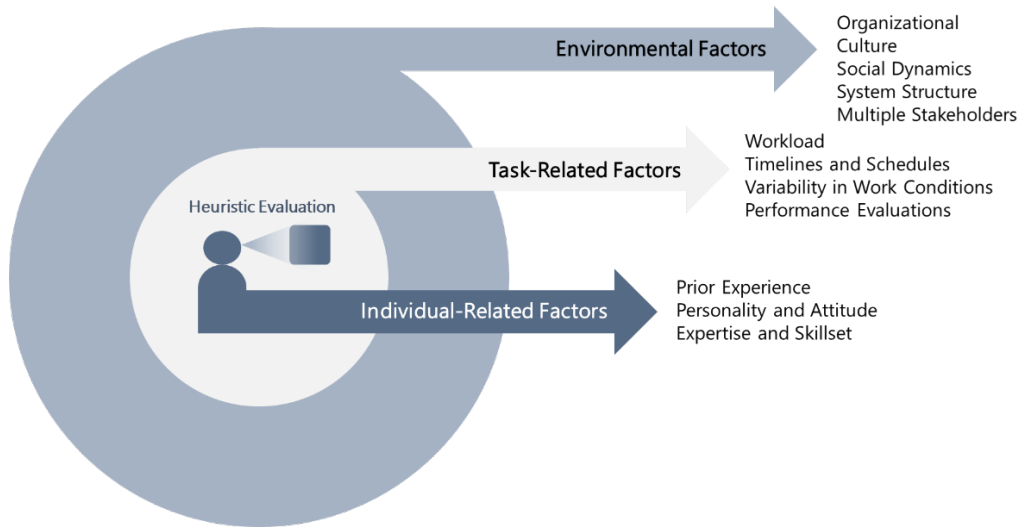


Figure 1: Factors influencing heuristic evaluation (HE).

experience related to it. So, the evaluator did not have the mental model to capture those kinds of issues,” illustrates that a diverse range of experiences can be crucial in spotting unique or specific usability issues.

However, while extensive familiarity with a product can lead to more nuanced and insightful evaluations, it also introduces the risk of developing cognitive blind spots, where certain aspects may be overlooked due to routine exposure. As TM-5 commented, “this could in some ways narrow the evaluators’ perspective.” The deep familiarity gained through long-term interaction with a product can undoubtedly enhance the efficiency and depth of evaluations, allowing experts to identify subtle issues or areas of improvement. However, it can also lead to cognitive biases, such as confirmation bias, where evaluators may become less open to alternative perspectives or innovative solutions. Therefore, it is essential to strike a balance between leveraging the depth of experience and avoiding tunnel vision.

On the other hand, varied experiences, perspectives, and skill levels among evaluators can also lead to communication difficulties within the organization. As TL-4 explains, “obvious gaps in the quality of HE results can be seen, because there are differences in the experience, thresholds, and perspectives of different evaluators. It is challenging to make this criteria or threshold to be the same for everybody.” Meanwhile, some experts mentioned that diversity could also provide a broader, more comprehensive view of the product, uncovering issues that might be overlooked by a more homogeneous team. The key lies in effective management of this diversity, particularly through the development of communication strategies that encourage open dialogue and mutual understanding.

In essence, the individual characteristics of evaluators bring both strengths and challenges to HE. Recognizing and balancing these aspects is key to developing more adaptive and effective HE methods.

3.2.2 Task-related Factors. HE is often more complex than it appears, influenced by factors like heavy workloads, tight deadlines,

and evolving product development timelines. As TM-5 noted, “HE depends on many factors; sometimes evaluation requests come before the product is fully developed, requiring us to evaluate incomplete features.” This dynamic requires evaluators to adapt their criteria and heuristics to the product’s state, emphasizing flexibility.

Furthermore, the demand for efficiency from tight schedules drive evaluators to develop personalized agile strategies. As TM-4 observed, “Experts conduct HE while managing other tasks simultaneously, so they tend to adapt different strategies to keep up” and TM-5 shared, “So, I focus only on UI changes to save time.” Another interesting point that we have found is that, when conducting HE, performance is frequently measured by the number of critical issues identified. Because of this, it creates a subtle pressure that makes the evaluators feel compelled to identify issues as a proof of diligence, as TM-5 observed, “when you conduct HE, usually your performance is measured by how many issues you find, especially critical issues, so what happens is that, evaluators are pushed themselves to find an issue, because if they can’t it seems that they did not work as hard”. This approach can inadvertently induce biases, particularly confirmatory bias, where evaluators may focus on identifying issues to demonstrate their effectiveness, rather than objectively assessing whether the UI genuinely has problems.

3.2.3 Environmental Factors. During our interviews, we uncovered a range of environmental factors that influence how evaluators conduct HE. Interestingly, factors such as organizational culture, social dynamics, and system structure greatly influence evaluators’ approaches.

A key observation was the impact of ‘social rules’ within organizations, particularly the desire to maintain status quo and harmonious relations with colleagues. This often leads evaluators to modify their approach to HE, in which they adapt a higher threshold for HE, focusing only on raising urgent and critical issues in order to avoid adding unnecessary workload to the team. As TL-4 mentioned, “The more issues you raise, the more burden you and your team will have.” This demonstrates their high regard for in-group

harmony, a feature common across many corporate cultures. While this approach may preserve team cohesion, it also risks underreporting issues that, though not immediately urgent, could have significant long-term usability consequences.

On a similar note, evaluators are sometimes reluctant to raise issues that they feel the other stakeholders would not agree with. Communicating the usability issues identified through HE to other stakeholders may sometimes prove difficult, especially because in a large corporation there are a lot of stakeholders involved: *“The thing here is that there is a lot of stakeholders. There are a lot of departments that are involved for a single product, things get more complicated.”* (TM-5). When conducting HE, the evaluators are not only required to identify usability issues, but also persuasively convey their findings to ensure stakeholder buy-in. TL-4 echoes this sentiment: *“[...] for issues that is identified through HE, it is difficult to raise it as it is, without conducting extra user tests, which can sometimes be unnecessary.”* Hence, for critical issues, experts usually conduct a separate user study to validate and complement their HE findings.

Interdepartmental communication adds another layer of complexity. TM-5 noted, *“With many stakeholders involved, it becomes difficult to raise issues without additional user testing, even if unnecessary.”* Organizational hierarchies further affect HE, as higher-ranking individuals’ opinions often carry more weight. While this can prioritize strategic goals, it may also overshadow evaluators’ practical insights. In addition, TM-2 highlighted the diffusion of responsibility [17] in large organizations: *“There are teams that have overlapping functions, so its easy to overlook some aspects of evaluation”* This narrowed scope may streamline workloads but risks overlooking minor usability issues that cumulatively degrade the user experience.

3.3 UX Practitioners’ Perspective on HE

There was broad consensus among participants that HE is a highly useful and essential method in the usability evaluation process. As TL-3 explained, *“HE is effective, especially in the early stages when you need to identify problems quickly. To make the feature more stable somehow.”* HE allows practitioners to uncover a high number of issues in a short time, which is particularly valuable in large tech organizations managing multiple products under tight timelines.

However, participants unanimously agreed that HE, while critical, should not be the sole method used in usability evaluations. TM-3 noted, *“It is efficient, but it is not a self-contained method. There is a lot of subjectivity in it. I would say it is necessary to conduct HE, but it should be the beginning of every user research or evaluation.”* Similarly, TL-4 emphasized its complementary role: *“HE is an essential part of the research process, but not something to take at face value. It’s efficient for identifying problems, but not necessarily for finding solutions or designing experiences.”* Participants characterized HE as a method best suited for identifying obvious usability issues (“big boulders”) that require quick fixes, which expedites product development. However, for ambiguous or specific issues, pairing HE with other methods, such as user testing, is essential.

A commonly noted limitation of HE is its inherent subjectivity. Different evaluators may interpret the severity or importance of issues differently, leading to inconsistent findings. TM-1 observed,

“Sometimes there are issues that one evaluator considered a problem, but others did not. For these ambiguous cases, we conduct further user testing and research to confirm our findings.” TM-3 described their organization’s process: *“We usually do HE first to identify potential problems. If issues are vague or require further validation, we conduct user tests. If more investigation is needed, we opt for large-scale quantitative analysis.”* These remarks highlight HE’s role as a starting point for identifying usability issues, guiding subsequent research to ensure objectivity and depth.

Participants also discussed challenges in using HE to persuade stakeholders, particularly in large organizations. TM-3 remarked, *“Results from HE can be difficult to communicate with others to make them understand the underlying issue.”* TM-4 added, *“You cannot persuade someone that something is an issue without visible data or evidence.”* While HE is effective at uncovering usability problems, it often lacks the persuasive power of methods that generate concrete evidence or user feedback. As a result, practitioners frequently supplement HE with additional methods to support their findings and build stakeholder buy-in.

4 Discussion

This case study laid the groundwork for HCI professionals to begin uncovering the realities of HE in practice. Findings from this research highlight the dynamic nature of HE in practical settings, revealing a divergence from idealistic methodologies and underscoring the influence of individual, task-related, and environmental factors.

Contrary to the belief that HE requires a formal set of heuristics and structured approaches in order to be effective, our study reveals a different story. The general practice of HE, as we found, has shifted to less structured methods, with UX practitioners increasingly relying on their intuition, personal experience, and knowledge of the product, moving away from traditional general heuristic sets. Moreover, due to the efficiency demands of the environment, formal procedures such as cross-evaluations, are rarely implemented in regular HE. This suggests that standard recommendations in the literature, such as an increase in the number of evaluators, the use of structured guidelines and checklists, and the adherence to a set of heuristics, may not always be optimal, especially in environments where efficiency is paramount.

This trend echoes the original informal essence of HE as proposed by Nielsen and Molich [11], but with a contemporary twist: evaluators now employ customized heuristics, unique to their experiences and knowledge. This finding challenges the notion of a universal set of heuristics and underscores the role of individual expertise in shaping evaluation methods. It reflects a broader trend in psychology where heuristics are seen as innate cognitive tools that guide decision-making, often operating subconsciously based on learned patterns and experiences [7].

While this approach may appear to be too subjective and potentially controversial in academic circles, we believe that this is just how it is. This is how we believe heuristics work. It reflects the inherent nature heuristics as mental shortcuts, not bound by strict set of rules but shaped by experience and contextual understanding. We recommend that future researchers delve deeper into the cognitive underpinnings of HE and examine the core skills needed for

effective assessment. This perspective challenges the recent focus on reducing HE's subjectivity and building structured methodologies and suggests a shift towards leveraging evaluators' knowledge and decision-making skills to enhance evaluation productivity.

Moreover, our study highlights the importance of considering a broader range of factors in HE, including individual characteristics, task-related conditions, and environmental contexts. These factors, often overlooked in HE literature, play a significant role in shaping the evaluators' judgments and decisions. One key insight from our research is the substantial impact of group dynamics and corporate culture on HE processes. For example, this study illustrates how evaluators' decisions and approaches are shaped by their position within the organizational structure. Decision-makers often have a more significant influence on the evaluation outcomes, which can both enrich and skew the process. In addition, adherence to social rules and nature of system structures (e.g., performance appraisals), do not only influence the evaluators' approach to HE but also affect their willingness to raise issues and attitude during the evaluation, revealing a complex interplay between individual judgments and collective norms. On top of this, tight schedules, high workload, and variability in working conditions, also affects how evaluators conduct HE, usually by adjusting their evaluation thresholds and thoroughness. These influencing factors however, should not be seen in a negative light. As this only reflects the reality of the practical world. This is something that should be acknowledged in the community: these environmental factors should not only be regarded as constraints that needs to be changed or improved, but rather it is the inevitable reality to where we need to base our judgments from. This notion encourages HCI professionals to incorporate a more comprehensive view of organizational culture and interpersonal dynamics into HE strategies.

The insights from this research provoke a rethinking of HE, suggesting that it should be approached as part of a comprehensive usability evaluation strategy, integrated with other methods and tailored to the specific needs of the setting. For UX practitioners, we recommend that instead of trying to make HE more comprehensive and systematic like other formal usability evaluation methods, one must strategically leverage its strengths to adapt to agile workflows. We must acknowledge that HE has its own strengths and weaknesses. For example, while HE is effective at quickly identifying major usability issues, its limitations—such as its subjectivity and lack of comprehensiveness—can be addressed by pairing it with more formal methods, such as user testing or quantitative analysis. We need to learn to leverage its strengths for our own advantage and address the weaknesses by combining it with other evaluation methods. We believe that by understanding that HE is a useful informal supplementary tool, it would pave the way for a more holistic and practice-oriented approach to HE.

For HCI researchers, future studies should explore how expert evaluators intuitively apply heuristics, identifying patterns that could inform more effective training and refinement of HE methods. Research should also focus on context-aware HE frameworks that adapt to industry-specific needs, particularly in AI-driven interfaces, enterprise UX, and emerging digital interactions. Additionally, examining organizational and social factors—such as corporate culture, team structures, and decision-making hierarchies—can

provide deeper insights into why some HE findings are overlooked or deprioritized.

Future researchers may wish to consider the perspective presented in this research and incorporate the findings of this study into their approach to the investigation of HE. However, it is important to note that this study was conducted within a single organization. Therefore, the generalizability of the results is limited. In order to gain richer insights, further research across different organizations is recommended. In addition, partnerships between industrial and academic researchers could also provide valuable insights into balancing theoretical best practices with practical constraints. Such collaborations can lead to more holistic and effective approaches to HE, tailored to the nuanced demands of different settings.

5 Conclusion

This case study, conducted within a large-scale organization, reveals a divergence between theoretical HE norms and real-world practices. While literature often emphasizes structured, multi-evaluator approaches, our findings show a preference for unstructured, experience-driven, single-expert evaluations in practical settings. Moreover, this research highlights the often-overlooked impact of individual, task-related, and organizational factors on HE processes.

We advocate for the development of flexible, context-aware HE models that integrate structured rigor with the adaptability of unstructured approaches. Such models would enable practitioners to leverage their expertise and intuition while addressing the practical demands of usability evaluations. Future research should delve deeper into the cognitive and organizational dimensions of HE to create frameworks that align with real-world needs.

In conclusion, this study challenges traditional HE frameworks and calls for adaptive approaches that reflect practical realities. By acknowledging the role of individual expertise and environmental factors, we can enhance the accuracy and efficiency of HE. This shift will empower UX teams to create user-centered products that succeed in complex, fast-paced environments.

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